HACKCULT '26 · HC26-GEDA01 · GREEN ENERGY & SUSTAINABILITY

LiFE-OS

Your electricity bill, turned into a footprint, savings, a solar plan and community action.

An interactive Mission LiFE citizen engagement application for the Gujarat Energy Development Agency.
Next.js · TypeScript · Supabase PostgreSQL

1 · THE PROBLEM

Awareness is high. Daily action is not.

Friction

Carbon calculators ask dozens of questions, so most people quit before seeing a number.

No link to rupees

Households respond to a lower bill. Kilograms of CO_2 alone rarely change a habit.

No path to rooftop solar

Interest stalls at "is it worth it for my roof?" with no honest estimate and no clear next step.

Who it serves: residential consumers, local communities, and state administrators monitoring regional conservation.

2 · THE SOLUTION

All four required features, working end to end

Personalised footprint tracker

One-minute onboarding from last month's units. Baseline, monthly trend, travel and LPG add-ons, and a visible calculation.

Nudges and micro-tasks

Suggestions from the user's own bills, the time of day and the home's equipment, each explaining why. Tasks have time windows and a weather rule, enforced on the server.

Gamification and social impact

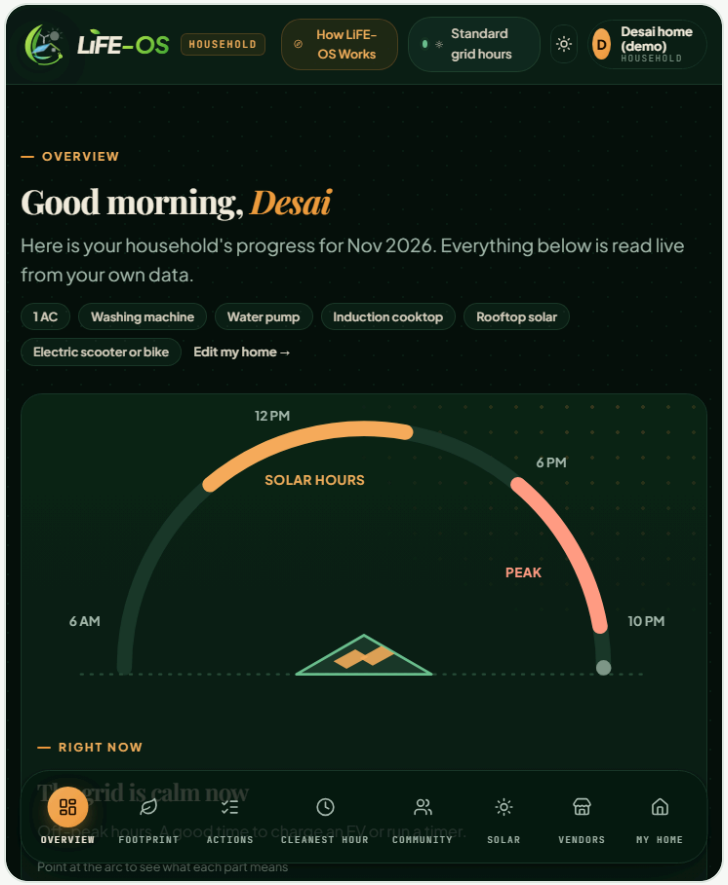
Points, streaks, levels, shared local and state-wide goals, opt-in leaderboards with trust tiers, and a "Verified citizen" badge for admin-approved photo proof.

Actionable resource hub

Rooftop solar estimate with ranges and assumptions, the steps to apply, and a map-based vendor directory that an administrator approves.

Plus: three roles (household, installer, GEDA administrator), an admin area map with privacy suppression, a bill scanner, and "Cleanest Hour" scheduling.

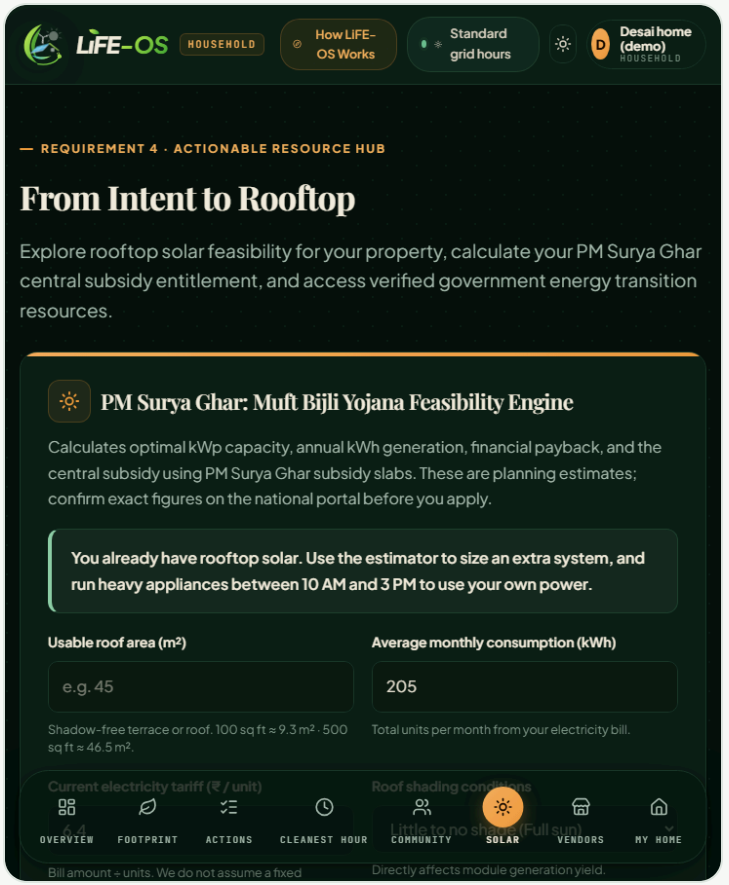
Real screens, real data flow



Dashboard: the day as a Sun Arc, footprint, points



Actions: time windows, weather rule, photo proof

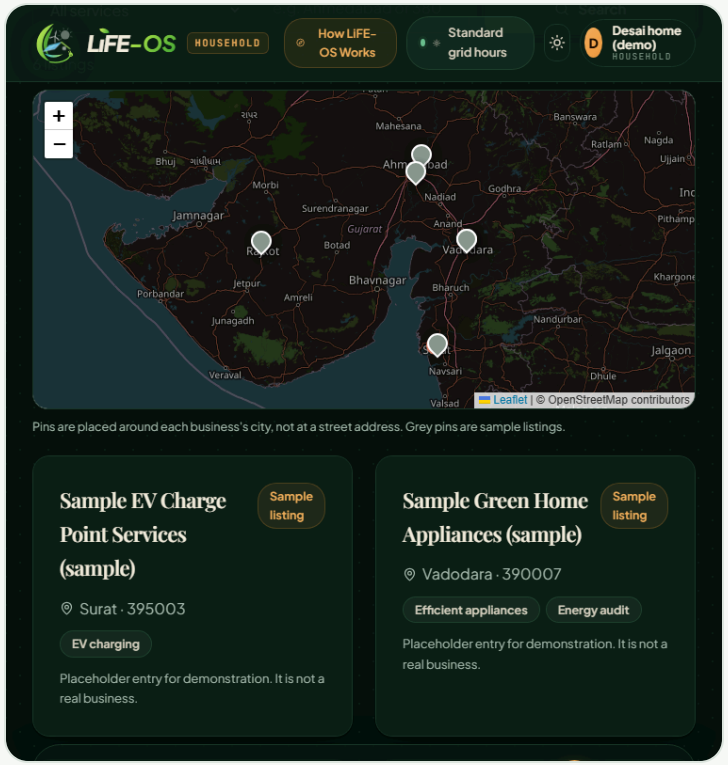


Resource hub: solar estimate with assumptions

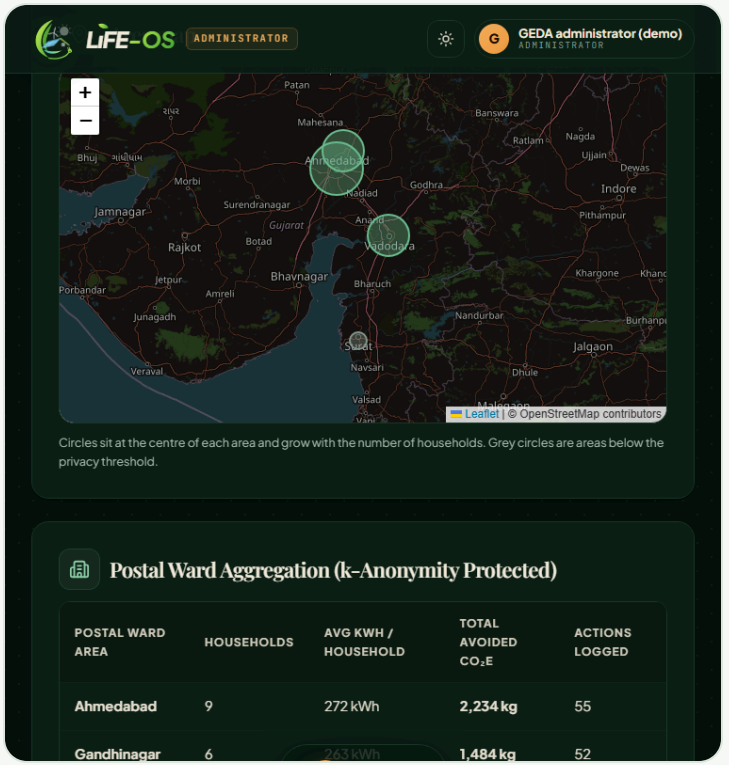
Three roles, maps and a fair way to earn trust



Welcome: sign in or open a demo household, installer or administrator



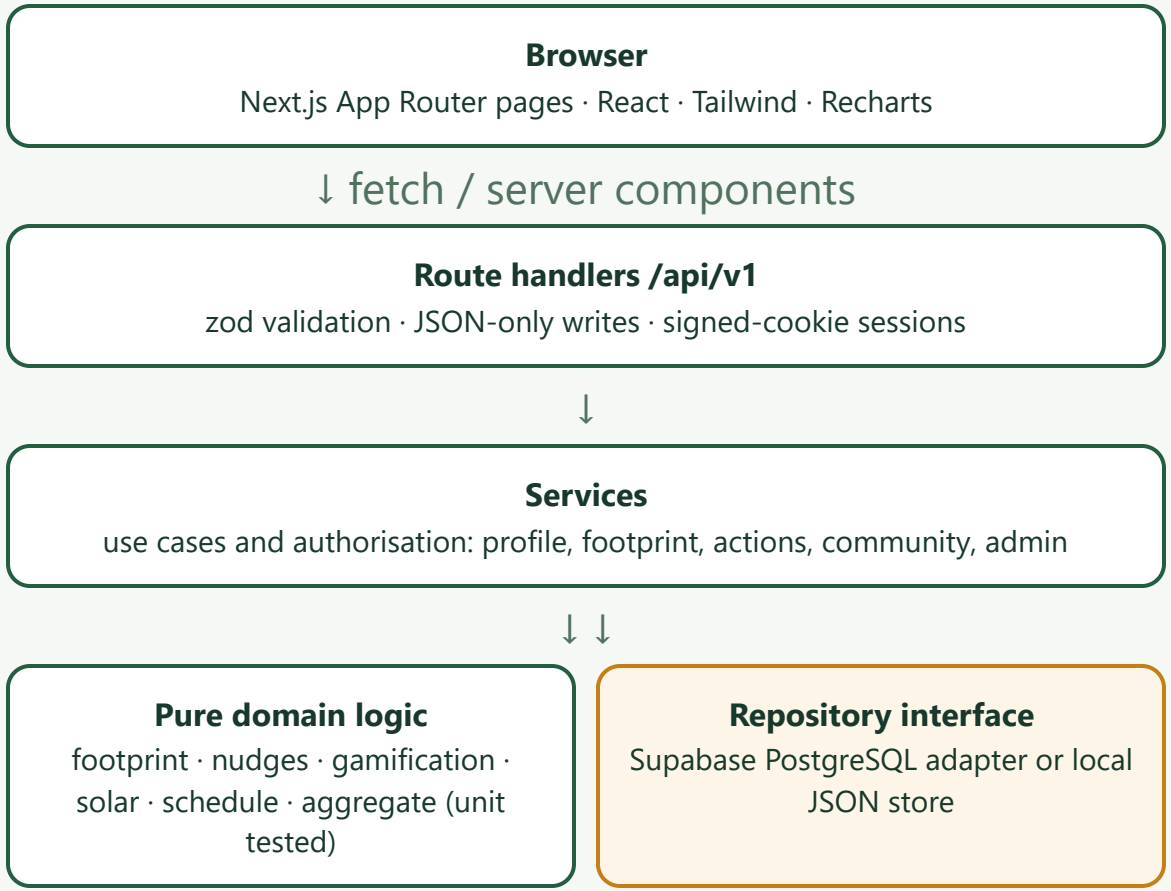
Vendor directory on a map, approved by an administrator



Administrator: area map and table, small groups hidden

4 · ARCHITECTURE

Layered, with one persistence boundary



Why it is built this way

- Business rules are pure functions, so they are testable and portable.
- Swapping the database touches one interface, not the app.
- Authorisation lives on the server, never in a hidden menu link.
- Zero-setup demo mode: the same code runs without any cloud account.

5 · DATABASE SCHEMA

Eight tables, constraints that enforce the rules

TABLE	KEY COLUMNS
app_users	id, role, email, password hash, pincode, home and equipment, leaderboard_opt_in, is_demo
footprint_snapshots	user_id, period, electricity_kwh, bill_amount_inr, total_kg, factor_version
task_completions	user_id, task_id, dedupe_key, completed_on, points, impact_kg
challenges	region_prefix, metric, target, starts_on, ends_on
challenge_members	challenge_id, user_id
vendors	name, services, pincode, status (pending, approved, rejected)
bill_proofs	user_id, period, consumer fingerprint, readings, units
task_proofs	user_id, task_id, status; the photo is deleted after review

Integrity in the database

- **unique(user_id, dedupe_key)** makes a double reward impossible, even under concurrent requests.
- **unique(user_id, period)**: saving a month again replaces it.
- **factor_version** on every snapshot keeps old estimates reproducible when factors change.
- **Row level security on, no public policy.** The publishable key can read nothing; only the server can.

6 · TECH STACK AND DECISIONS

One deployable app, no glue services

Next.js 15 · React 19 · TypeScript strict · Tailwind 4 · Recharts · zod · Supabase PostgreSQL · Vitest

143 unit tests cover footprint maths, nudges, solar sizing, points rules, task time windows and the weather rule, streaks, challenge progress, admin suppression, bill reading and checks, photo-proof storage and IST date handling. Lint, typecheck and production build all pass. Live on Render with Supabase PostgreSQL.

Key decisions

- Next.js route handlers instead of a separate API server: one deploy, one language.
- Email and password accounts (salted scrypt hashes) with signed cookie sessions, and three roles enforced on the server. Verified email through Supabase Auth is the next step.
- India Standard Time for every "today", so streaks and limits match users' days.

7 · WHY IT STANDS OUT: TRUST BY DESIGN

Honest numbers, fair play, private by default

Bills that must add up

The scanner reads a bill on the device. Readings times the multiplier must equal the billed units, and one bill cannot serve two accounts.

Explain every nudge

Each suggestion states why it fired and how its saving was estimated.

Fair points

One reward per window, a daily cap, and server-side scoring. Big actions need an admin-approved photo, deleted after review.

Privacy

Leaderboards are opt-in. Admins see only areas with 5+ households, grouped by 3 PIN digits.

Every estimate shows its assumptions, and unverified inputs are labelled as such in the interface.

Built to grow, honest about today

Scales because

- Stateless, pure domain logic scales horizontally on serverless.
- Idempotent writes enforced by database constraints.
- Indexes on per-user history and PIN-prefix grouping.
- Next: admin and leaderboards move to SQL views (`GROUP BY left(pincode,3) HAVING count(*) >= 5`) with paginated ranking.

Limitations today

- Emission factors are indicative, not the current CEA release.
- Solar subsidy figures are unverified assumptions.
- Cleanest Hour is a modelled curve, not live grid data.
- Everyday tasks are self-reported. Three big ones need a photo that a person approves: a human look, not proof the appliance is real.
- The bill scanner proves the numbers are consistent, not that the bill is genuine.
- No satellite roof tracing or AI assistant. Smart-meter confirmation and email verification are planned, not built.

From prototype to statewide programme

Next

- Verified email or phone OTP and password reset, with per-user RLS
- Confirm scanned bills with the DISCOM or smart-meter data
- Roof outline on the map with PVGIS yield

Then

- DISCOM tariff slabs from official GERC orders
- Hourly grid intensity series for Cleanest Hour
- Gujarati and Hindi interfaces

The ask

Pilot LiFE-OS with a DISCOM and a few wards: verified tariffs, real bill data, and measured savings.